Project Report: Iteration 2

Sean Shea & Jacob Ford
The RPG Platformer (Still Working Title)

Goals

Create basic, operational RPG and Platformer Scenes. Not yet linking the two gameplay forms, just operating on an individual basis. While not fine tuned yet, both scenes should give a sample basis of how player/enemy interactions work.

Tasks

Provided is the list of tasks originally planned from the initial write-up, as well as details regarding division of responsibility and task status.

- RPG
    - Basic Battle System
        - Allow Player to attack and Enemy to attack, and for HP to update accordingly.
        - *Responsibility:* Sean
        - *Status:* Complete
    - Visuals for HP, Mana, Enemy HP.
        - *Responsibility:* Sean
        - *Status:* In Progress
    - Battle Start systems
        - Allow a battle to be started from whatever state the player and enemies are in.
        - *Responsibility:* Sean
        - *Status:* In Progress
    - Battle End Systems
        - Cement design for leaving battle sequences in various ways
        - *Responsibility:* Sean
        - *Status:* Moved to future Iteration
- Platforming
    - Basic player and enemy physics
        - Create basic enemy type and have them interact using the in game physics
        - *Responsibility:* Jacob
        - *Status: Complete*

    - Simple Enemy AI
        - Create basic enemy movement and responses to in game events
        - *Responsibility:* Jacob
        - *Status: Complete*
    - Attack features
        - Player and enemy both have a way to interact with each other affecting stats (health) to show functionality
        - *Responsibility:* Jacob
        - *Status: Complete*
    - Combat Transition Details
        - Figure out means through which the platformer will detect and activate the start of a battle sequence
        - *Responsibility:* Jacob
        - *Status: Complete*
    - Basic Level Design
        - Create a basic level for the player to traverse to a goal in.
        - *Responsibility:* Jacob
        - *Status: Moved to future iteration*
- General
    - Working Placeholder animations
        - Find and implement art and animations for game elements
        - *Responsibility:* Sean, Jacob
        - *Status: Moved to future Iteration*

Completed Work

RPG

Basic Battle System: A basic battle system is in place with turns and different actions for the player. The Player can either FIGHT (do a small amount of damage to the enemy), SKILLS (at the moment just uses SP (Magic Point Equivalent) and does double damage), or ITEMS (Heals the player).

Visuals for HP, Mana, Enemy HP: Visuals for the Player's HP and SP has been implemented, and visuals for Enemy HP has been started, though I ran into some problems implementing it so it is not complete yet

Platforming

Basic Player and enemy Physics: Both the player and the basic created enemy are affected by gravity and scene elements, as well as each other.

Simple Enemy AI: There is now a basic enemy type that will just move back and forth on the screen, damaging the player as it comes into contact with it. This simple enemy has a stat block that can be generalized to all enemy entities as well as a moveset that can be updated to provide more complicated features.

Attack Features: The player now has a sword that will damage the enemy when it comes in contact with it, and the enemy does damage upon touch. In addition, to prevent excessive repeated damage, the enemy is knocked back upon a hit and both are knocked back when the player is hit.

Combat Transition Details: The concept of how to determine when the platformer will transition to a battle atmosphere has been decided. It was determined that for the sake of prioritizing a working concept above all else that the method of instantiating a battle should be simplified to start, adding additional features later. To start, this will be done simply by checking how many enemies are within a specified distance upon user request, and starting a battle with the closest 3 from there. To be further developed and implemented in the next iteration.

Incomplete or Changed Work

RPG

Battle Start systems: A system to keep track of stats has been started, so far it's just a script that holds entity stats that is used for both player and enemies. In the future the plan is to use this script as a scriptable object and have it used to communicate stat changes between the RPG system and platformer system

Battle End Systems: Currently not started due to lack of time. Will use a similar system to the Start system.

Platforming

Basic Level Design: It was determined that this should be held off till all individual elements of the game are in working condition, which is not a reality quite yet. As such, we are currently just remaining mindful of what elements are needed for a complete level (platform interaction, player/enemy mechanics) to later be pieced together. As such, this will be implemented at a future iteration.

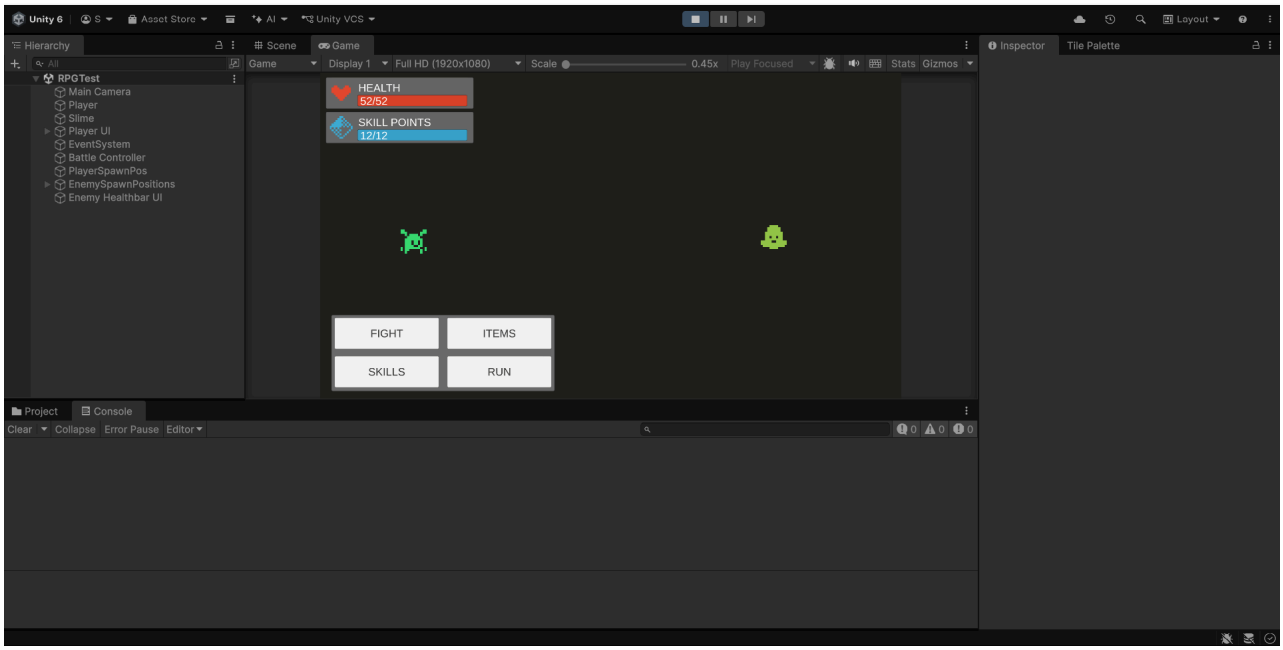
General

Working Placeholder Animations: It was discovered that we can do a large amount of the work without needing to have the final art or animations to make the mechanics operable, so it was determined to wait till everything in the game works at a base level before implementing specific visual design elements.

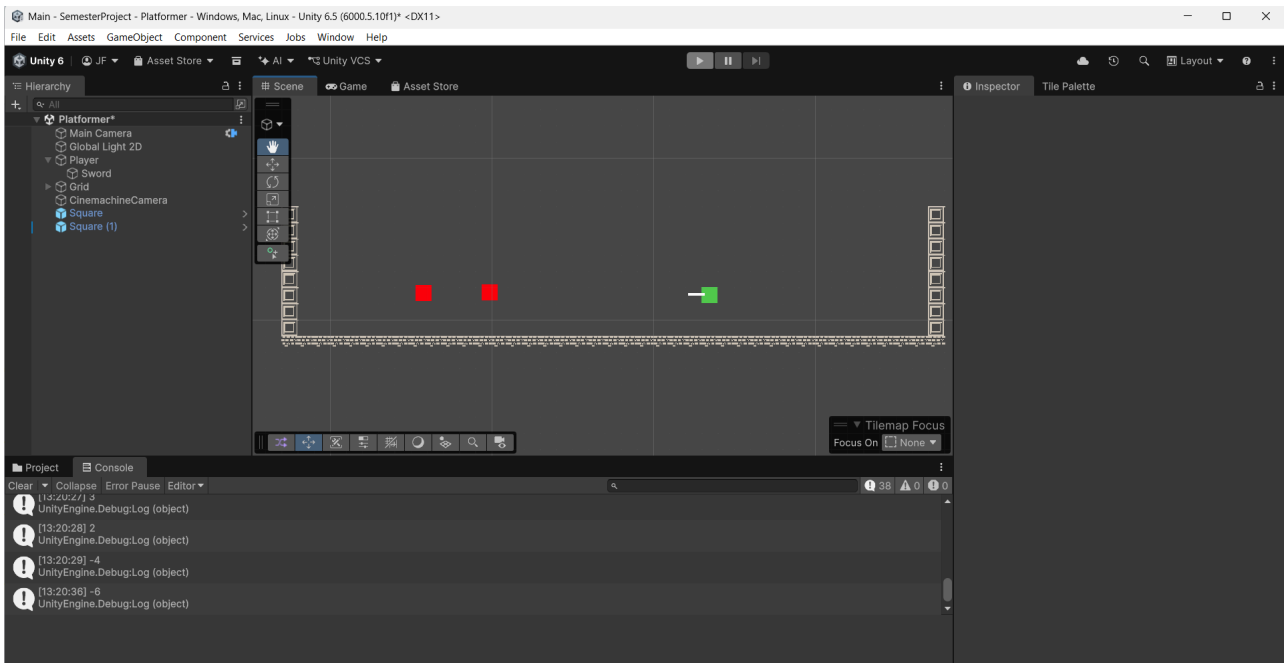
Changes From Initial Project Plan

No major changes from initial project plan, though there has been a slight degree of downscaling expectations and focus to essential elements.

Current Project Evidence



Unity environment running the RPG test Scene (See Iteration Two Elements/RPGExample.mkv in GitHub for a more detailed video).



Unity environment for the Platformer test scene (See Iteration Two Elements/PlatformerDemo in GitHub for a more detailed video look)

Development and Contribution Evidence

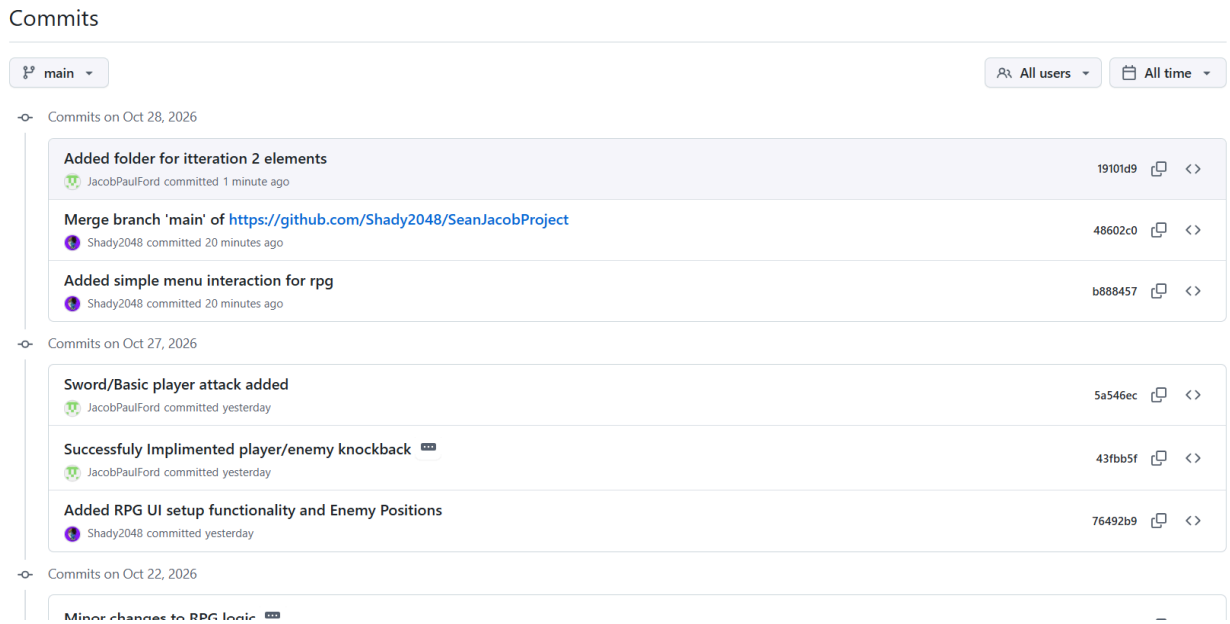
GitHub: https://github.com/Shady2048/SeanJacobProject

Note: GitHub names are as follows -

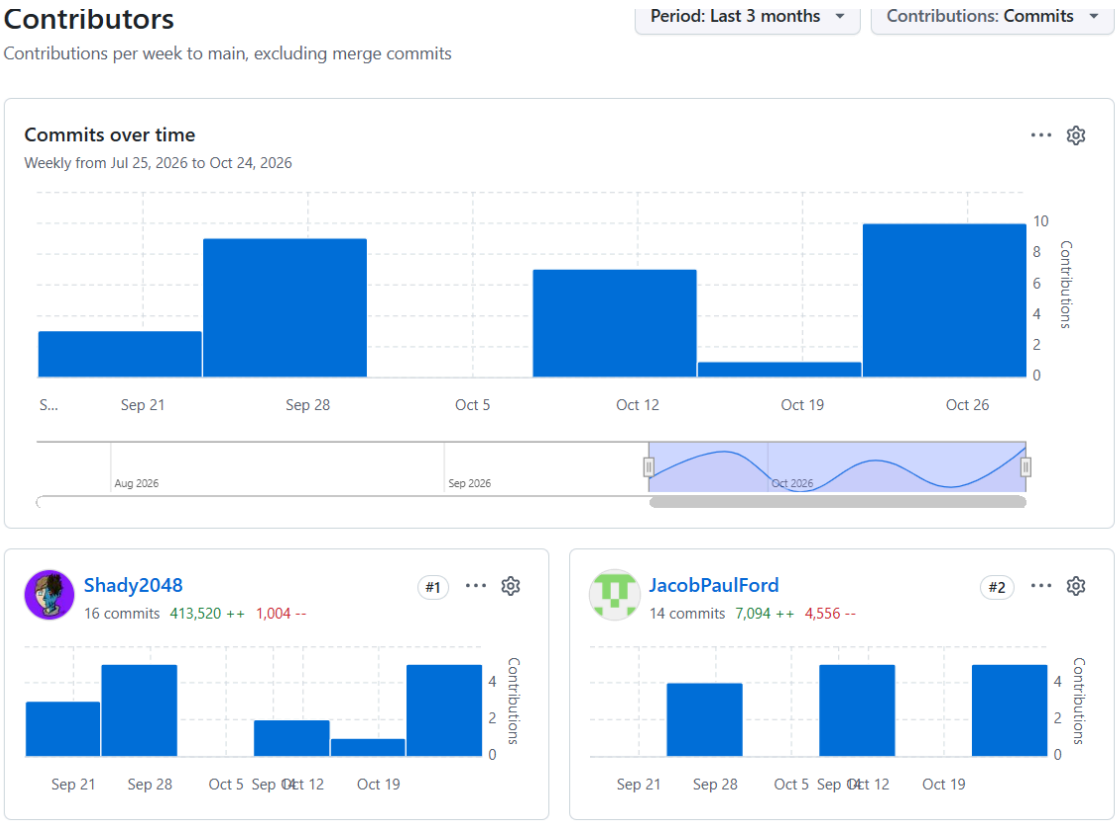
Jacob Ford - JacobPaulFord

Sean Shea - Shady2048

We uploaded the Unity Project to GitHub and are using it to allow both of us to work on the same project, where general contribution elements will be marked, and we are using Google Docs to manage our ideas and take notes on our thoughts to stay organized.



Commit Logs with Meaningful titles and comments are added for every commit to record changes. For a full list, please see the github repository for a complete list under “Iteration Two Elements\Commits · 10-28-2026”



Current statistics of commit contributions to the project.

Problems, Concerns, and Lessons Learned

We have begun to recognize we may have over-fantisized the scope of this project. It is still more than completable, but we have begun to recognize we need to better distinguish which features are actually necessary for a functioning game and which features are desired, but expendable should we not have the time to implement it before the project is due.

Goals for Iteration 3

General Idea - Add functionality to shift between the two systems, and allow the game to be playable from both sides.

- RPG
    - Finish up RPG Battle system
    - Enemy health bars implementation
    - Multiple Skill Attacks that use different amounts of SP
    - Add systems for leaving battles in multiple ways

    - Add better visual indicators for what is happening in the battle (Player feedback)
- Platforming
    - Add input support for starting battle transition
    - Add enemy variation (multiple types/stats)
    - Add goal trigger and death triggers
    - Add enemy death triggers
    - Polish player attack
- General
    - Add information flow support between scenes
    - Basic Starting scene, ending scene, and pause screen.
    - Add replayability